

# Supplementary Material: Early Black-Hole Growth Constraints from JWST

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The main paper reports required average Eddington ratios for 224 objects in its Balmer-line sample and 234 objects in the expanded sample. This supplement contains the additional constant-spin grid and the early-start extrapolation.

## S1 Growth equations

We use  $f_{\text{Edd}} = L_{\text{bol}}/L_{\text{Edd}}$ , radiative efficiency  $\epsilon$ , seed mass  $M_{\text{seed}}$ , and merger multiplier  $B_{\text{merge}}$ . The time available for growth is  $\Delta t = t(z_{\text{obs}}) - t(z_{\text{seed}})$ . At fixed efficiency, growth depends on the time-average  $\overline{f_{\text{Edd}}}$ :

$$M_{\text{BH}}(t_{\text{obs}}) = B_{\text{merge}} M_{\text{seed}} \exp \left[ \frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \Delta t}{t_{\text{Edd}}} \right], \quad (\text{S1})$$

$$\log_{10} \left( \frac{M_{\text{seed, req}}}{M_{\odot}} \right) = \log_{10} \left( \frac{M_{\text{BH}}}{M_{\odot}} \right) - \log_{10} B_{\text{merge}} - \frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \Delta t}{t_{\text{Edd}} \ln 10}. \quad (\text{S2})$$

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_{\Lambda}}} \operatorname{asinh} \left[ \sqrt{\frac{\Omega_{\Lambda}}{\Omega_m}} (1+z)^{-3/2} \right], \quad (\text{S3})$$

Here  $t_{\text{Edd}} = 0.45$  Gyr. The age relation uses  $H_0 = 67.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $\Omega_m = 0.315$  and  $\Omega_{\Lambda} = 0.685$  and neglects radiation. The main calculations begin at  $z_{\text{seed}} = 30$ . The extension to  $z = 3400$  below is a mathematical extrapolation of this age relation.

## S2 Efficiency prescription

For the illustrative spin-dependent maps, the ideal Kerr thin-disk efficiency is the ISCO binding efficiency ([J. M. Bardeen et al. 1972](#)):

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3r_{\text{ISCO}}}}, \quad (\text{S4})$$

$$r_{\text{ISCO}} = 3 + Z_2 - \operatorname{sgn}(a) \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \quad (\text{S5})$$

$$Z_1 = 1 + (1 - a^2)^{1/3} [(1 + a)^{1/3} + (1 - a)^{1/3}], \quad Z_2 = \sqrt{3a^2 + Z_1^2}. \quad (\text{S6})$$

The dimensionless spin is  $a = cJ/(GM_{\text{BH}}^2)$ , where  $J$  is the signed angular momentum relative to the disk, and  $r_{\text{ISCO}}$  is in units of  $GM_{\text{BH}}/c^2$ . The ideal efficiencies at  $a = -1, 0, +1$  are approximately 0.038, 0.057,

and 0.423, respectively, neglecting photon capture and stress at the ISCO. Logarithmic luminosity growth at supercritical inflow is motivated by advection and wind models (J. Poutanen et al. 2007). We use a simplified logarithmic relation with coefficient one and omit radial mass loss. For Eddington ratios above unity, the maps adopt

$$f_{\text{Edd}} = 1 + \ln \dot{m}, \quad \dot{m} = \frac{\dot{M}_{\text{in}}}{L_{\text{Edd}}/(\epsilon_a c^2)}, \quad \epsilon_{\text{eff}} = \begin{cases} \epsilon_a, & f_{\text{Edd}} \leq 1, \\ \epsilon_a f_{\text{Edd}} \exp(1 - f_{\text{Edd}}), & f_{\text{Edd}} > 1. \end{cases} \quad (\text{S7})$$

The logarithmic luminosity relation applies only for super-Eddington accretion. Each compatibility cell holds both the rate and its effective efficiency constant over the interval. The grid tests constant accretion states with fixed spin and omits relativistic slim-disc dynamics. For time-varying efficiency, growth depends on  $\int [(1 - \epsilon(t))/\epsilon(t)] f_{\text{Edd}}(t) dt$ , which requires the joint time dependence of efficiency and accretion rate.

Figure S1 gives the full scenario grid. The broad  $10^2$ – $10^6 M_\odot$  interval is stored under the original code key `pbh_origin_hypothesis`. This key labels the mass interval; a primordial interpretation requires additional assumptions about seed formation. The descriptive fractions use point masses and the inclusive interval criterion in Section S3, without error integration. Section S4 explains the conditional interpretation of this interval for pre-existing primordial black holes.

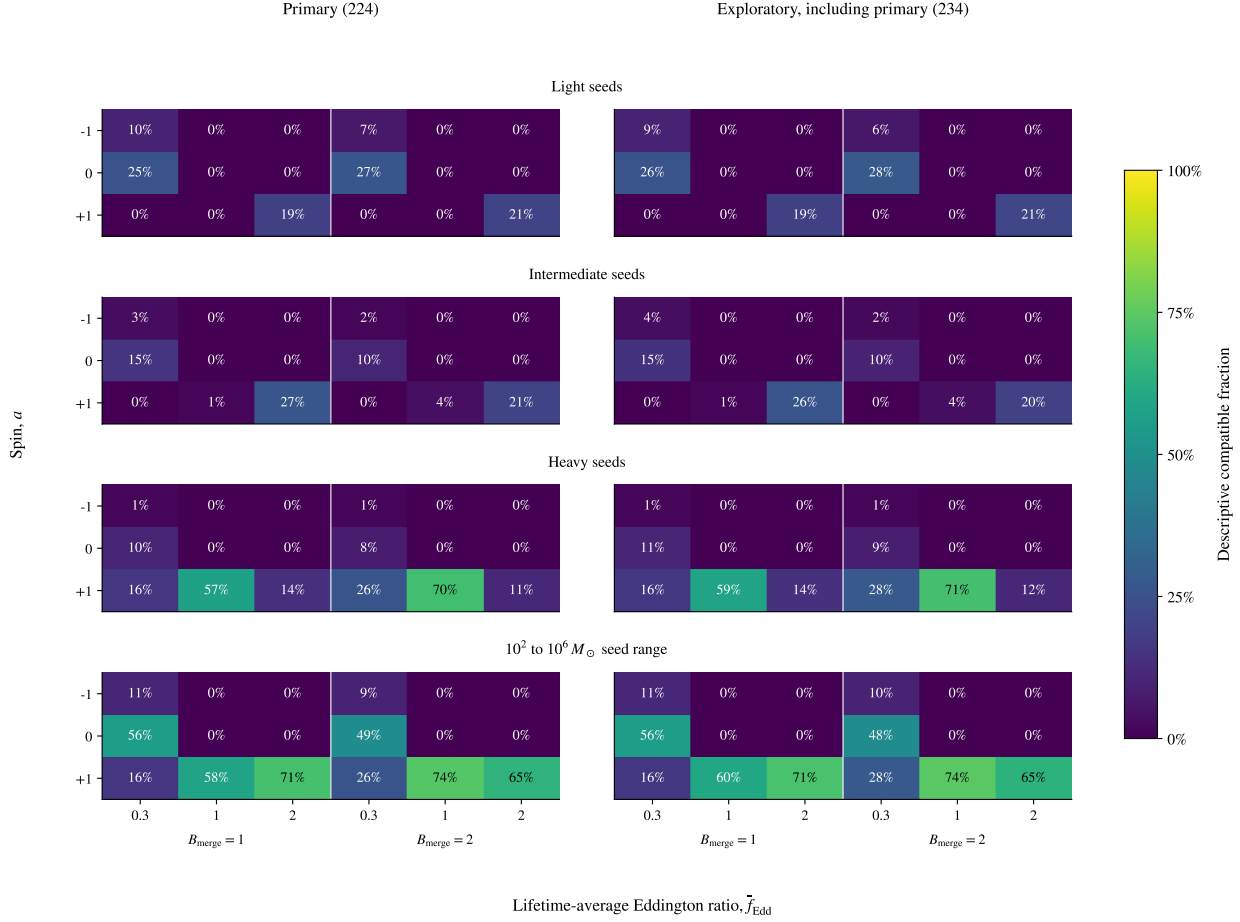
### S3 Compatibility across growth assumptions

For each assumed accretion rate, efficiency, starting redshift and merger contribution, we solve Eq. S2 for the initial seed mass that would grow to the observed black-hole mass. We then check whether that seed falls in each of four mass ranges:  $10$ – $100 M_\odot$  (light),  $10^3$ – $10^4 M_\odot$  (intermediate),  $10^4$ – $10^6 M_\odot$  (heavy), and  $10^2$ – $10^6 M_\odot$  (broad). We count an object as compatible with a range when some seed within it can reproduce the observed mass under these assumptions. The grid uses  $z_{\text{seed}} = 30$ , three spin/efficiency cases, merger multipliers 1 and 2, and average rates 0.3, 1 and 2. These maps use the supercritical efficiency prescription in Section S2.

The seed-mass ranges overlap and are tested separately, without probability weights. The broad range spans several possible seed-formation channels. If the seed mass needed to reproduce an object is smaller than the lower bound of a tested range, even the smallest seed in that range would grow larger than the observed black hole at the assumed accretion rate. That object is therefore excluded from the count for that range and set of growth assumptions. For each set of assumptions, we report the percentage of catalogue objects whose observed masses can be reproduced from a seed in the tested range. These percentages describe this literature sample. Estimating the corresponding percentages among all high-redshift black holes would require accounting for how each contributing study selected and detected its objects.

The primary and exploratory samples show similar compatibility patterns (Fig. S1). For heavy seeds at  $a = +1$  and  $f = 1$ , changing  $B_{\text{merge}}$  from one to two raises compatibility from 57 to 70 per cent in the primary sample (59 to 71 per cent inclusive). At higher accretion rates, even the smallest seed in a tested mass range can grow larger than the observed black hole. Such objects reduce the percentage reproduced from seeds in that range. These fractions use central mass estimates and the efficiency prescription in Section S2. They describe compatibility within the selected sample; estimating the frequency of each formation channel would require uncertainty propagation and a population selection model.

### Compatibility across growth assumptions



**Figure S1:** Constant-state compatibility fractions in the primary sample (left) and the inclusive exploratory sample (right). The exploratory sample includes all primary objects. Each column specifies merger multiplier  $B$  and fixed luminosity ratio  $f$ ; rows specify ideal spin-dependent efficiency, with the supercritical prescription in Section S2. Seed intervals overlap and have no assigned probability weights. The broad mass interval spans several possible seed-formation channels.

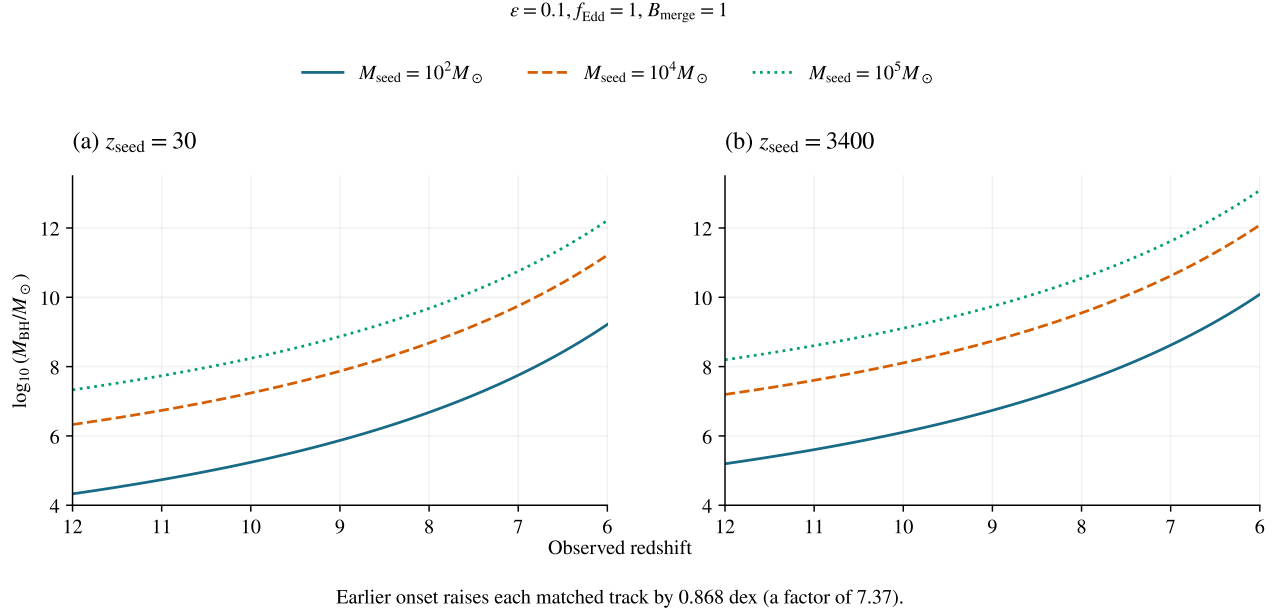
## S4 Earlier growth onset and primordial seed interpretations

The growth equations take the initial mass and subsequent accretion history as inputs. The seed-formation mechanism is left unspecified. A primordial black hole (PBH) already present at  $z = 30$  can therefore be represented by its effective mass at that epoch if it lies within the broad  $10^2$ – $10^6 M_{\odot}$  interval and follows the adopted subsequent growth prescription. This mass need not equal its primordial formation mass. This interval covers a specified range of effective masses; inferring a primordial origin requires a separate formation model. In P. Dayal (2024),  $z \simeq 3400$  marks matter–radiation equality and the start of the modelled evolution of PBHs that formed earlier.

Moving the start of the same constant-rate, constant-efficiency history from  $z = 30$  to 3400 adds  $\delta t = t(30) - t(3400) = 0.09990$  Gyr in our matter-plus-Lambda extrapolation. At fixed seed mass and merger multiplier, the resulting shift is

$$\Delta \log_{10} M_{\text{BH}} = \frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \delta t}{t_{\text{Edd}} \ln 10}. \quad (\text{S8})$$

For  $\overline{f_{\text{Edd}}} = 1$ , this is 0.868 dex (a factor of 7.37) at  $\epsilon = 0.1$ , or 1.589 dex (a factor of 38.85) at the ideal non-spinning thin-disc efficiency. The tracks shift upward by these amounts (Figure S2). At fixed observed mass and seed mass requiring positive accretion, the ratio of required average rates is  $[t(z_{\text{obs}}) - t(30)]/[t(z_{\text{obs}}) - t(3400)]$ ; the earlier start lowers the requirement by 6.5, 13.1 and 22.9 per cent at  $z_{\text{obs}} = 4, 7$  and  $10.603$ , respectively.



**Figure S2:** Growth from  $z_{\text{seed}} = 30$  (a) and  $3400$  (b), with matched seed masses and the fixed assumptions shown. Earlier onset adds 99.90 Myr and shifts each track upward by 0.868 dex. Here  $z_{\text{seed}}$  denotes the assumed start of accretion. The  $z = 3400$  comparison extrapolates the matter-plus-Lambda cosmology and assumes uninterrupted accretion. Radiation, gas supply, halo assembly and feedback are omitted.

These comparisons quantify the additional growth time. Applying them to PBHs requires an effective mass at the chosen starting epoch and a model for the subsequent gas supply and accretion history.

## References

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